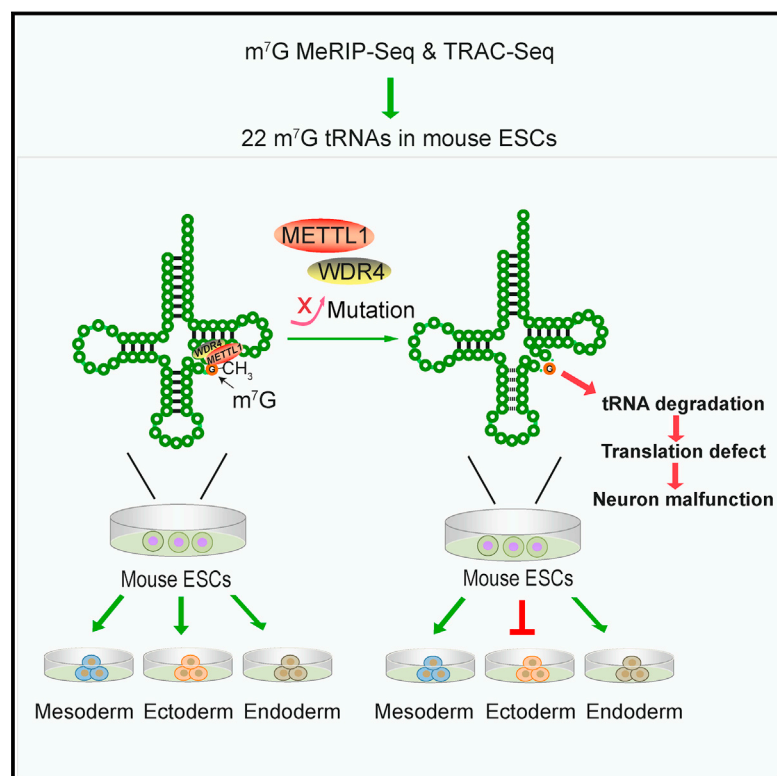


Mettl1/Wdr4-Mediated m⁷G tRNA Methylome Is Required for Normal mRNA Translation and Embryonic Stem Cell Self-Renewal and Differentiation

Graphical Abstract



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In Brief

Lin and Liu et al. developed two independent methods, MeRIP-seq and TRAC-seq, to profile the m⁷G tRNA methylome in mouse ESCs and revealed that Mettl1/Wdr4-mediated m⁷G tRNA methylome is required for normal mRNA translation and ESC self-renewal and differentiation.

Highlights

- Develop MeRIP-seq and TRAC-seq for mapping of m⁷G modification
- Reveal the m⁷G tRNA methylome in mammalian cells at single-nucleotide resolution
- Mettl1/Wdr4-mediated m⁷G tRNA methylome is required for normal mRNA translation
- Mettl1/Wdr4-mediated m⁷G tRNA methylome regulates ESC self-renewal and differentiation

Data Resource

GSE112670



Mettl1/Wdr4-Mediated m⁷G tRNA Methylome Is Required for Normal mRNA Translation and Embryonic Stem Cell Self-Renewal and Differentiation

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SUMMARY

tRNAs are subject to numerous modifications, including methylation. Mutations in the human *N*⁷-methylguanosine (m⁷G) methyltransferase complex *METTL1*/*WDR4* cause primordial dwarfism and brain malformation, yet the molecular and cellular function in mammals is not well understood. We developed m⁷G methylated tRNA immunoprecipitation sequencing (MeRIP-seq) and tRNA reduction and cleavage sequencing (TRAC-seq) to reveal the m⁷G tRNA methylome in mouse embryonic stem cells (mESCs). A subset of 22 tRNAs is modified at a “RAGGU” motif within the variable loop. We observe increased ribosome occupancy at the corresponding codons in *Mettl1* knockout mESCs, implying widespread effects on tRNA function, ribosome pausing, and mRNA translation. Translation of cell cycle genes and those associated with brain abnormalities is particularly affected. *Mettl1* or *Wdr4* knockout mESCs display defective self-renewal and neural differentiation. Our study uncovers the complexity of the mammalian m⁷G tRNA methylome and highlights its essential role in ESCs with links to human disease.

INTRODUCTION

tRNAs are subject to numerous posttranscriptional modifications, including methylation, which can control tRNA folding, stability, and function in mRNA translation (Chou et al., 2017; Liu et al., 2016; Phizicky and Alfonzo, 2010). Dysregulation of specific tRNA modifications has been linked to a range of genetic disorders and cancers (Kirchner and Ignatova, 2015; Torres

et al., 2014). Individual tRNA modifications have different functional roles, with several of the tRNA modifications in the anticodon region being essential for translation and growth in yeast, whereas the numerous tRNA modifications outside of the anticodon region are often nonessential and might regulate tRNA stability, folding, and/or function under certain growth conditions, including the stress response (Kirchner and Ignatova, 2015; Phizicky and Alfonzo, 2010).

*N*⁷-methylguanosine (m⁷G) at nucleotide 46 (m⁷G46) is one of the most prevalent tRNA modifications found in prokaryotes, eukaryotes, and in some archaea. In yeast, m⁷G46 is found in the variable loop region of 11 tRNA species (Sprinzl et al., 1998), including PheGAA, ValAAC, ValCAC, MetCAT, and LysTTT (Phizicky and Alfonzo, 2010), and the tRNA^{Phe} structure shows a C13–G22–m⁷G46 base triple interaction (Jovine et al., 2000). The m⁷G modification is also one of the few known modifications that introduces a positive charge or zwitterion onto the nucleobase. However, m⁷G is a nonessential tRNA modification in yeast. The yeast methyltransferase Trm8p and its essential co-factor Trm82p form a heterodimeric complex that catalyzes the S-adenosylmethionine (SAM)-dependent m⁷G modification of tRNA (Alexandrov et al., 2002; Leulliot et al., 2008). Depletion of Trm8, Trm82, or double mutations of Trm8 and Trm82 has no effect on tRNA expression and does not affect yeast growth under normal culture conditions (Alexandrov et al., 2005, 2006). However, the Trm8 or Trm82 yeast mutants have increased sensitivity to high temperature and show growth defects due to the rapid decay of tRNA ValAAC under heat stress (Alexandrov et al., 2006).

In human, mutations in the gene encoding *WDR4* (WD repeat domain 4), the ortholog of yeast Trm82p and a component of the human m⁷G tRNA methyltransferase complex, causes impaired tRNA m⁷G modification and a distinct form of microcephalic primordial dwarfism characterized by facial dysmorphism, brain malformation, and severe encephalopathy with seizures (Shaheen et al., 2015; Trimouille et al., 2018). *WDR4*, located at



human chromosome 21q22.3, is also a candidate gene for some of the Down syndrome phenotypes, including intellectual disability caused by trisomy of this chromosomal region in human patients (Michaud et al., 2000). Indeed, *Wdr4* is one of a handful of candidate genes whose overexpression in mouse influences learning and memory in a model of Down syndrome (Pereira et al., 2009). This suggests that the m⁷G tRNA modification might have more important physiological functions in mammals compared to yeast. In addition, the m⁷G tRNA methyltransferase METTL1 (methyltransferase like 1; ortholog of Trm8) was reported to be regulated by the protein kinase B (PKB) and ribosomal S6 kinase (RSK) under growth factor stimulation (Cartledge et al., 2005) and influences cancer cell viability and sensitivity to 5-fluorouracil (Barbieri et al., 2017; Okamoto et al., 2014). The more severe phenotypes and physiological functions associated with dysregulation of m⁷G tRNA modifications in humans suggest differential evolutionary requirements for m⁷G tRNA modifications between mammals and yeast. However, the full extent of m⁷G tRNA modification has not been examined in a mammalian system, and the molecular and cellular mechanisms underlying the severe developmental consequences of defective m⁷G tRNA modification due to METTL1/WDR4 deficiency remain unknown.

In this study, we developed two independent methods, m⁷G methylated tRNA immunoprecipitation sequencing (MeRIP-seq) and m⁷G tRNA reduction and cleavage sequencing (TRAC-seq) to profile the global m⁷G tRNA methylome in mouse embryonic stem cells (mESCs). Our data identified the global m⁷G tRNA methylome that is essential for the proper translation of cell cycle genes and genes associated with brain abnormalities in mESCs. In addition, knockout of *Mettl1* or *Wdr4* resulted in impaired mESC self-renewal and neural lineage differentiation capacity, consistent with the severe brain malformations seen in human patients with WDR4 mutations. Our study uncovers the complex tRNA m⁷G methylome in mammals and highlights the essential molecular and cellular role of *Mettl1*-*Wdr4* and m⁷G modification in regulation of stem cells and disease.

RESULTS

Global Profiling of tRNA m⁷G Modification by m⁷G MeRIP-Seq

To study the tRNA m⁷G methylome in mammalian cells, we first knocked out the tRNA m⁷G methyltransferase *Mettl1* using CRISPR/Cas9 in mESCs (Figures 1A, S1A, and S1B). Analysis of RNA extracted from control and *Mettl1* knockout ESCs using an anti-m⁷G Northwestern assay, whereby extracted total RNA samples were resolved on a denaturing polyacrylamide gel, transferred to a nylon membrane, and blotted using α -m⁷G antibody, revealed the specific depletion of the m⁷G signal in tRNAs in the *Mettl1* knockout (KO) cells but did not affect m⁷G-modified rRNAs (Figures 1B and S1C). We further generated a mutant of the m⁷G methyltransferase METTL1 by mutating the active site (amino acids [aas] 160–163; LFPD to AFPA) and found that expression of the wild-type (WT) METTL1 cDNA in the KO ESCs rescued the m⁷G modification on tRNAs whereas expression of the catalytic mutant METTL1 (Mut) did not (Figures 1C and 1D). We further confirmed by northern blot that MeRIP per-

formed using the anti-m⁷G antibody specifically enriched the known m⁷G-modified tRNAs, but not the U6 small nuclear RNA (snRNA) control. Moreover, *Mettl1* depletion specifically abolished the enrichment of m⁷G-modified tRNAs in the anti-m⁷G MeRIP (Figure 1E).

To examine m⁷G modifications globally, we developed a small RNA m⁷G MeRIP-seq methodology using small RNAs (<200 nt) isolated from KO and control ESCs to identify the global m⁷G modification in tRNAs. Because the extensive modifications on tRNAs can block the reverse transcription, we adopted a recently developed strategy of using the purified bacterial demethylase AlkB to remove tRNA methylation to achieve more efficient and accurate tRNA sequencing (Figure 1F; Cozen et al., 2015; Zheng et al., 2015). This revealed that AlkB treatment of input small RNAs dramatically increased the proportion of tRNA reads in the sequencing library (Figure 1G, left four samples). Sequencing of the anti-m⁷G MeRIP enriched and the input small RNAs revealed that m⁷G MeRIP-seq strongly enriched certain tRNAs, and this enrichment was abolished in the *Mettl1* KO cells (Figure 1G). Analysis of m⁷G MeRIP-seq data uncovered the widespread m⁷G modification of tRNAs in mESCs. In addition to the several known m⁷G tRNAs, we identified a total of 21 tRNAs that contain m⁷G modifications (Figures 1H and 1I; Table S1). Several of the newly identified m⁷G tRNAs were confirmed by m⁷G MeRIP and northern blotting (Figure 1J). Given that, in yeast, only 11 m⁷G tRNAs were reported, our data revealed a more complex m⁷G tRNA modification landscape in mESCs.

Single-Nucleotide Resolution Mapping of tRNA m⁷G Modifications by m⁷G TRAC-Seq

We next used a chemical cleavage method to confirm the newly identified m⁷G modifications on tRNAs. RNAs were subjected to reduction with sodium borohydride (NaBH₄) and aniline-induced cleavage (Wintermeyer and Zachau, 1975; Zueva et al., 1985), and the cleaved tRNA fragments were analyzed by northern blot using specific tRNA probes (Figures 2A and 2B). Our data revealed that, after chemical treatment, cleaved tRNA fragments could be detected in the RNA samples from WT cells, but not the *Mettl1* KO cells, supporting the specificity of the chemical method in the cleavage of m⁷G-modified RNAs. We next aimed to apply this chemical method for single-nucleotide resolution identification of m⁷G sites by adaptor ligation and sequencing. 18S rRNA was tested because it has a known m⁷G site and is long enough for PCR-based cloning and sequencing (Figure S2A). The total RNA sample from mESCs were first treated with NaBH₄ and aniline and then an RNA adaptor was ligated to the 5' end of the cleaved 3' fragments, and RT-PCR was performed using primers to amplify the region that contains the adaptor-cleaved fragment junction site (Figures S2A and S2B). Sanger sequencing of the PCR products revealed the chemical treatment followed by adaptor ligation can specifically detect the known m⁷G site on G1639 of mouse 18S rRNA (Figure S2C).

To perform the single-nucleotide resolution mapping of global m⁷G modifications in tRNAs, we developed the m⁷G site-specific tRNA reduction and cleavage-sequencing (TRAC-seq) method. The isolated small RNAs were first treated with AlkB that removes many methylation modifications but does not

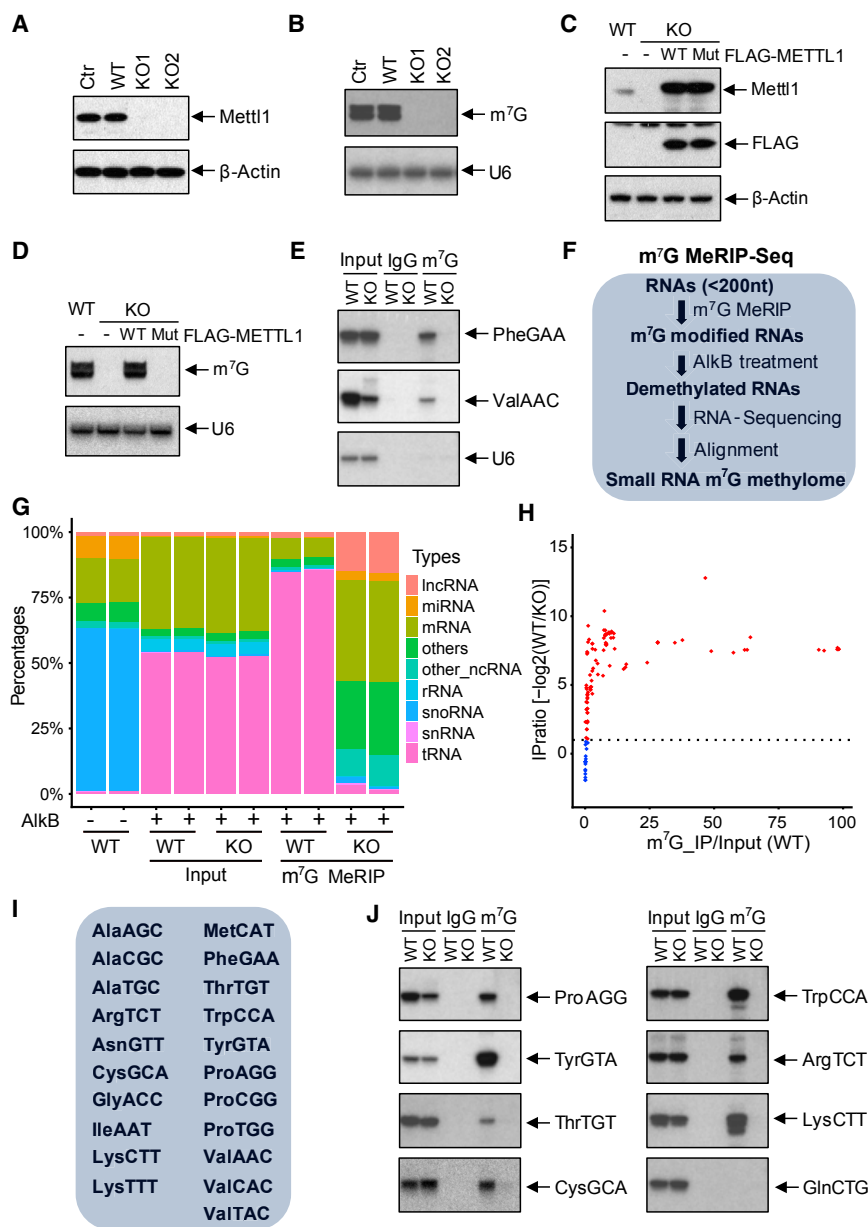


Figure 1. Global Profiling of tRNA m⁷G Modifications by m⁷G MeRIP-Seq

(A) Western blot of the control and Mettl1 KO samples with indicated antibodies. (B) Anti-m⁷G Northwestern blot of m⁷G modifications (upper panel). RNA samples were separated on a TBE-urea gel and then transferred to nylon membrane for western blotting using anti-m⁷G antibody. Northern blot of U6 snRNA was used as a loading control (lower panel). (C) Western blot of the METTL1 rescue samples with indicated antibodies. (D) Anti-m⁷G Northwestern blot of m⁷G modifications (upper panel) and northern blot of U6 snRNA (lower panel). (E) Small RNA northern blot of the anti-m⁷G IP products with indicated probes. (F) Schematic diagram of small RNA m⁷G MeRIP-seq to identify the m⁷G methylome. Small RNAs were isolated and treated with AlkB to remove modifications and then used for cDNA library construction. (G) Population of small RNA mapping in m⁷G MeRIP-seq. Samples without AlkB treatment (AlkB-) were used as controls. n = 2. (H) Scatterplot of tRNA relative enrichment of tRNA in m⁷G MeRIP-seq. Relative enrichment was calculated by comparing the IP/input ratio in the WT samples to the KO samples. (I) List of m⁷G-modified tRNAs identified by m⁷G MeRIP-seq. (J) Confirmation of m⁷G-modified tRNAs by anti-m⁷G IP and northern blot with indicated probes. No m⁷G modification was identified in GlnCTG, which was used as a control.

affect m⁷G, and AlkB-treated RNA was subjected to chemical reduction and cleavage conditions, as described above. A cDNA library was then constructed with adaptor ligation and PCR amplification. The sequence reads obtained from deep sequencing of these libraries were aligned to a small RNA database to identify new cleavage sites induced by chemical treatment (Figure 2C). Our m⁷G TRAC-seq data revealed that chemical treatment resulted in the site-specific initiation of reads after the known m⁷G46 site of tRNA PheGAA in WT mESC cells, but not in Mettl1 KO cells (Figure 2D). We also found site-specific cleavages in other known m⁷G tRNAs, as well as those newly identified by m⁷G MeRIP-seq (Figure 2D). To identify m⁷G cleavage sites in tRNAs globally, we calculated the cleavage score by comparing the ratio of the number of reads that initiated from a

specific site to the number of reads across that site in the treated and control samples (Figure 2E). We found a site-specific increase of cleavage score in the m⁷G sites, which allowed us to identify global m⁷G modifications in tRNAs at single-nucleotide resolution (Figure 2F). Analysis of the m⁷G TRAC-seq identified a total of 22 tRNAs that contain m⁷G modifications (Figures 2F and S3), which includes all of the 21 m⁷G-modified tRNAs identified in our m⁷G MeRIP-seq as well as IleTAT (Figure 2G). IleTAT expression was very low in the input samples and was below the level of detection in the m⁷G MeRIP-seq (not shown). Overall, our tRNA m⁷G MeRIP-seq and m⁷G TRAC-seq reproducibly identified the tRNA m⁷G methylome in mESCs and uncovered a considerably more complex tRNA m⁷G modification landscape in mammals than had been previously found in yeast tRNAs.

Evolutionally Different m⁷G tRNA Modification and Function in Mouse and Yeast

The fact that more tRNAs contain m⁷G modification in mouse than in yeast suggested the potential evolutionary differences in tRNA m⁷G modification and function between different species.

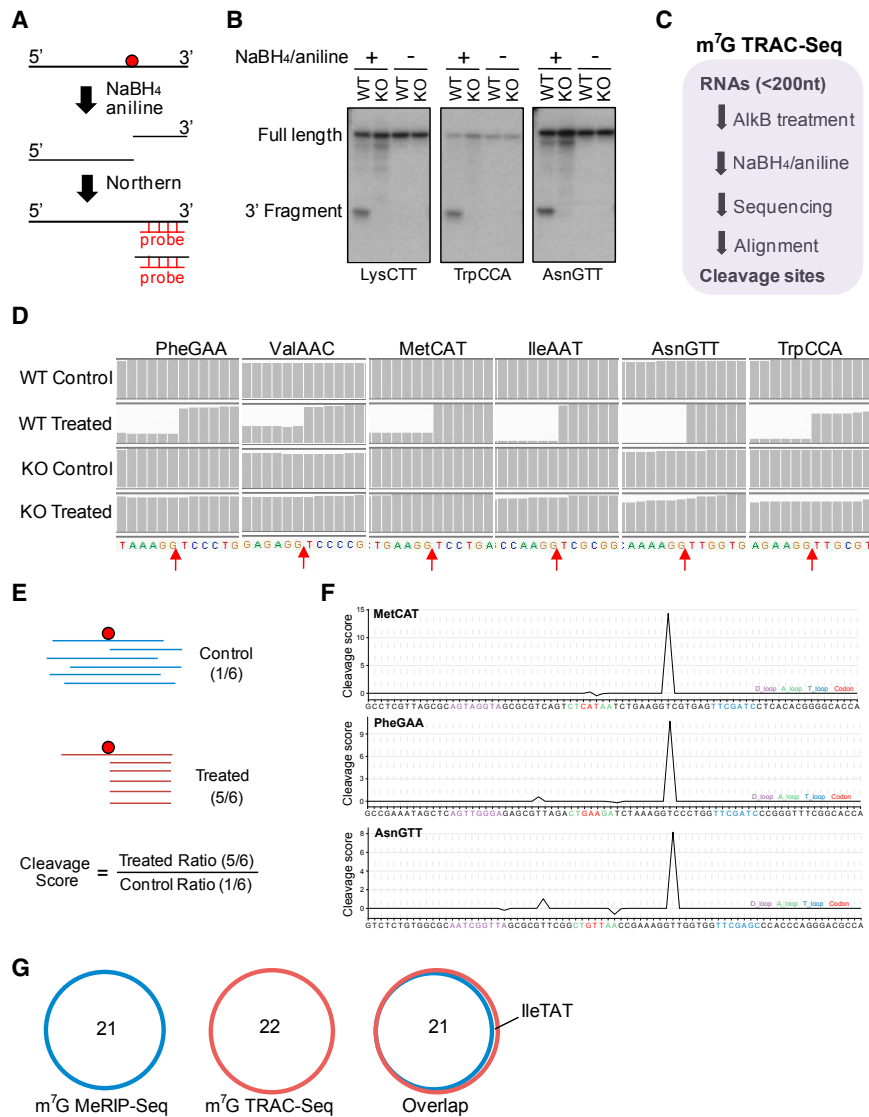


Figure 2. Single-Nucleotide Resolution Mapping of m⁷G Modifications by m⁷G TRAC-Seq

(A) Schematic diagram of tRNA reduction and cleavage at m⁷G sites. RNA samples were treated with NaBH₄ and aniline to induce the site-specific cleavage of RNA at m⁷G-modified guanosines. (B) Northern blot of the chemical-treated RNA samples with indicated probes. (C) Schematic diagram of the m⁷G TRAC-Seq. (D) Representative images of reads alignment to indicated tRNAs in the Integrative Genomics Viewer (IGV) using m⁷G TRAC-seq data. Chemical treatment resulted in a site-specific decrease of reads at m⁷G-modified position. n = 2. (E) Schematic diagram of the method to calculate the cleavage score at individual positions using TRAC-seq data. (F) Representative images of cleavage scores of indicated tRNAs. Pictures show a specific increase of cleavage score at the m⁷G sites. (G) Overlap of m⁷G-modified tRNAs identified in m⁷G MeRIP-seq and m⁷G TRAC-seq.

We further determined whether the function of m⁷G tRNA modification is different between mouse and yeast. In yeast, depletion of the m⁷G tRNA methyltransferase Trm8 or its partner Trm82 has no effect on the expression level of m⁷G containing tRNA and does not affect yeast growth under normal culture conditions at 28°C but does induce sensitivity at elevated growth temperature of 37°C (Alexandrov et al., 2005, 2006). In mouse, we found that KO of *Mettl1* resulted in expression changes in the majority of m⁷G-modified tRNAs under normal culture conditions at 37°C. The mean expression of 6 tRNAs, including ValTAC, ValAAC, ValCAC, ProCGG, ProAGG, and IleAAT, decreased more than 2-fold in

Analysis of cleavage sites in m⁷G TRAC-seq identified a “RAGGU” m⁷G modification motif, in which the second guanine is modified (Figure 3A). We next studied the mechanisms underlying the differences in m⁷G-modified tRNAs between mouse and yeast. Currently, there are 60 types of tRNA in mouse and only 42 types of tRNA in yeast annotated in the Genomic tRNA Database (Chan and Lowe, 2016). Comparison between the mouse and yeast tRNA genes revealed that several RAGGU-motif-containing tRNAs, for example, GlyACC, are present in mouse, but not in yeast (Figure 3B). Moreover, although some tRNAs are common in both mouse and yeast, their sequences in the V-loop have slightly changed during evolution. For example, tRNAs ArgTCT and AsnGTT contain the “AAGGU” motif in the V-loop in mouse, although in yeast and other lower organisms, the corresponding sequences are “AAGAU” (Figure 3C). Overall, these data suggest that tRNA types and sequences within the V-loop of tRNA have evolutionarily changed, resulting in a more complex and widespread m⁷G tRNA methylome in mammals.

Mettl1 KO cells, although expression of several other tRNAs was more modestly decreased (Figures 3D–3F). Surprisingly, CysGCA expression is increased when *Mettl1* is depleted (Figures 3D–3F). Overexpression of the WT *METT1*, but not the catalytic dead mutant, can rescue the expression of the ProAGG and ValAAC m⁷G tRNAs in the *Mettl1* KO cells (Figure 3G). Determination of tRNA aminoacylation by acid urea PAGE followed by northern blot revealed that depletion of tRNA m⁷G has little effect on tRNA aminoacylation (Figure S4). Taken together, our results revealed that the m⁷G tRNA modification is widespread and is essential for the proper expression of m⁷G-modified tRNAs in mammalian cells under normal growth conditions.

Mettl1 Depletion Causes Ribosome Pausing at m⁷G-tRNA-Dependent Codons and Is Required for the Translation of Cell Cycle Genes

To study the function of *Mettl1*-mediated m⁷G tRNA modifications, we first performed polysome profiling to determine the

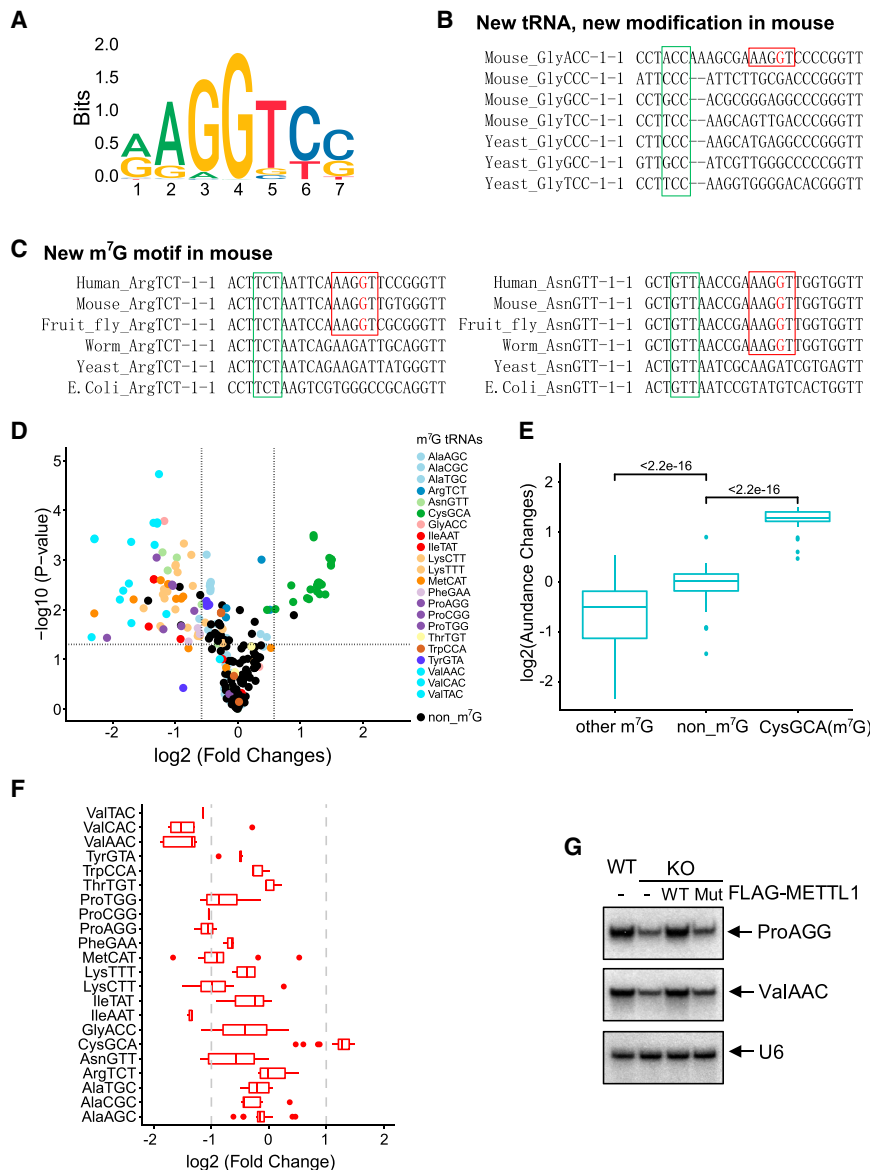


Figure 3. Evolutionary Sequence Conservation and Function of m⁷G tRNA Modifications

(A) Sequence motif in the m⁷G sites identified by m⁷G TRAC-seq in mouse cells.

(B) Sequences of Gly-tRNAs in mouse and yeast. GlyAAC tRNA exists in mouse, but not in the yeast. Green box labels the anticodon, and red box indicates the m⁷G motif.

(C) Sequences of ArgTCT and AsnGTT tRNAs in different species. Green box labels the anticodon, and red box indicates the m⁷G motif.

(D) Scatterplot of tRNA expression levels in the Mettl1 KO and control cells. Each dot represents one tRNA gene. x axis indicates the changes of tRNA expression levels; y axis indicates p value.

(E) Mann-Whitney test on the expression of the m⁷G and non-m⁷G tRNAs.

(F) Expression profile of the 22 m⁷G-modified tRNAs. Each boxplot shows the expression of a tRNA type that was calculated from the combined expression of all the tRNA genes for the same tRNA type.

(G) Small RNA northern blot of the METTL1 rescue samples with indicated probes. U6 snRNA was used as a loading control.

role of m⁷G tRNA modifications in regulation of global translation. Our result revealed that depletion of Mettl1 causes a reduction of polyribosome peak (Figure 4A), suggesting that global translation is decreased in the Mettl1 KO cells. We further performed ribosome footprinting to identify the specific genes with altered translation in Mettl1-deficient cells (Table S2). Our ribosome footprinting identified ribosome-protected fragments (RPF) that are located mainly in the coding DNA sequence (CDS) as expected (Figures S5A–S5C). Analysis of the translation efficiency (TE) and codon frequency revealed that the mRNAs with decreased translation efficiency have a significantly higher frequency of the codons decoded by the m⁷G-modified tRNAs. On the other hand, the mRNAs with increased translation efficiency have a lower frequency of m⁷G-tRNA-dependent codons (Figures 4B and 4C). Comparison of ribosome occupancy in the Mettl1 KO and WT cells revealed that depletion of Mettl1 results in the

increased ribosome pausing at m⁷G-tRNA-dependent codons in the charged tRNA binding sites (A sites), and as control there is little change in the ribosome occupancy in the A+1 site, suggesting that the Mettl1-mediated tRNA modification is essential for efficient codon recognition during mRNA translation (Figure 4D). Moreover, ribosome occupancy in Mettl1-deficient cells was overall more pronounced at the A sites compared to the P sites, consistent with perturbation at the codon recognition step (Figures S5D and S5E). However, certain other codons that are decoded by m⁷G-modified tRNAs, including AUA(I) and GUA(V), showed increased ribosome occupancy at the P sites, suggesting that loss of m⁷G for the corresponding tRNAs might impede ribosome translocation (Figure S5D). The codon occupancy for the m⁷G-modified tRNAs (except for CysGCA) is significantly higher than the codon occupancy for the non-m⁷G-modified tRNAs (Figure S5F). Consistent with the increased expression of CysGCA tRNA, the codon occupancy for CysGCA tRNA decreased compared to occupancy on the non-m⁷G tRNAs decoded codons (Figure S5F). We also tested for the correlation between A site occupancy changes and tRNA abundance changes and revealed that decreased tRNA abundance is significantly associated with increased A site occupancy (Figure S5G). Taken together, these data further support the essential role of m⁷G tRNA modification in regulation of tRNA level and mRNA translation.

Gene ontology analysis of human phenotypes associated with the TE-decreased genes upon Mettl1 KO revealed that those

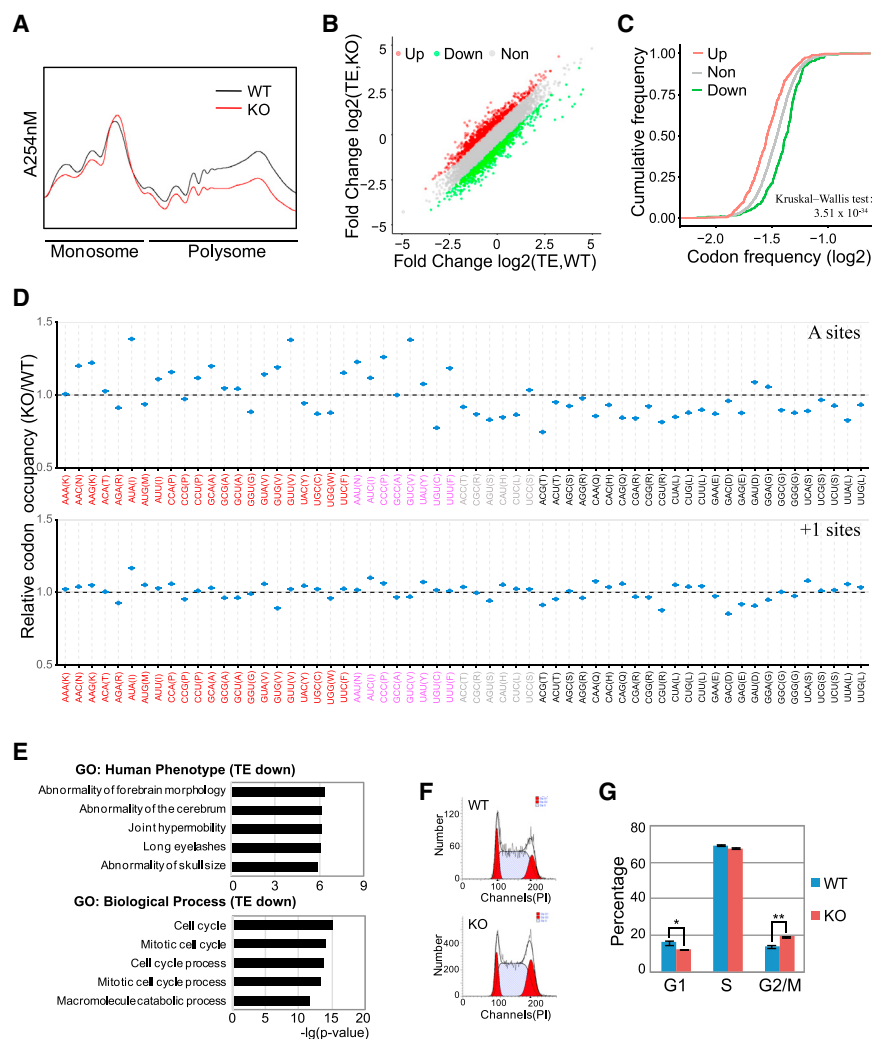


Figure 4. Mettl1 Depletion Causes Ribosome Pausing at m⁷G tRNA-Dependent Codon Containing Cell Cycle and Disease Genes

(A) Polysome profiling of the Mettl1 KO and WT cells. (B) Scatterplot of translation efficiency (TE) in the Mettl1 KO and WT cells. TE was calculated by dividing the ribosome-protected fragments (RPF) signals to the input RNA-seq signals. (C) Codon frequency in the CDS region of the genes with increased TE (up), decreased TE (down), and other genes (non) in the Mettl1 KO cells. (D) Ribosome occupancy at individual codon at A sites and A+1 sites. Data were presented as the relative ribosome footprinting signals from Mettl1 KO and WT cells. The codons are separated into four groups based on the modification and expression status of their corresponding tRNAs: codons decoded by m⁷G tRNAs (red); codons decoded by m⁷G tRNAs by wobble effect because of the not detected levels of their corresponding tRNAs (pink); codons decoded by non-m⁷G tRNAs by wobble effect because of the not detected levels of their corresponding tRNAs (gray); and codons decoded by non-m⁷G tRNAs (black). (E) Gene ontology analysis of function enrichment in human phenotype and biological process using the TE downregulated genes upon Mettl1 KO. (F and G) Cell cycle analysis (F) and quantification (G) of Mettl1 KO and WT cells. n = 3; *p < 0.05; **p < 0.01.

genes are selectively enriched in abnormality of forebrain morphology, cerebrum, and skull size (Figure 4E), which is consistent with the brain malformation and microcephaly described in WDR4 mutated patients. Moreover, the genes with decreased TE are associated with cell cycle progression (Figure 4E), suggesting the role of Mettl1-mediated m⁷G tRNA methylation in regulation of the cell cycle. Indeed, cell cycle analysis revealed that, in the Mettl1 KO cells, the percentage of cells in the G2 phase is significantly increased (Figures 4F and 4G), suggesting that depletion of Mettl1 results in impaired G2/M transition. Overall, our data uncovered the function of Mettl1-mediated m⁷G tRNA methylation in regulating the translation of disease and cell-cycle-related genes.

Mettl1 Is Required for ESC Self-Renewal

Considering the widespread m⁷G tRNA modifications and their role in regulating translation of disease and cell cycle genes in the mESCs, we decided to study the requirement of Mettl1-mediated m⁷G tRNA modifications in mESC biology. Depletion of Mettl1 resulted in decreased expression of stem cell markers Nanog and KLF4 (Figures 5A and 5B). Moreover, consistent

with the impaired cell cycle progression, the Mettl1 KO cells proliferate more slowly than the control cells (Figure 5C). Depletion of Mettl1 resulted in smaller colony size and impaired colony-formation ability (Figures 5D–5F). In addition, alkaline phosphatase (AP) staining revealed a significant decrease of the AP-positive population in the Mettl1 KO cells (Figures 5G and 5H), suggesting that Mettl1-depleted mESCs have defective self-renewal and may undergo partial differentiation. Overall, the above data suggested that Mettl1 is essential for normal ESC self-renewal.

We further performed RNA sequencing analysis to study the global gene expression profiles in the Mettl1 KO cells. Our data identified 174 upregulated and 174 downregulated genes in the Mettl1 KO mESCs (Figure 5I; Table S3). Gene ontology analysis revealed that the upregulated genes function in regulation of animal organ morphogenesis and development (Figure 5J). On the other hand, the downregulated genes are associated with negative regulation of multicell organismal process and cell differentiation (Figure 5K). Real-time qRT-PCR was used to confirm the gene expression changes in the Mettl1 KO cells (Figures 5L and 5M). Taken together, our results identify a critical role of Mettl1-mediated m⁷G tRNA methylome in maintaining ESC self-renewal and suggest that Mettl1 might influence ESC differentiation.

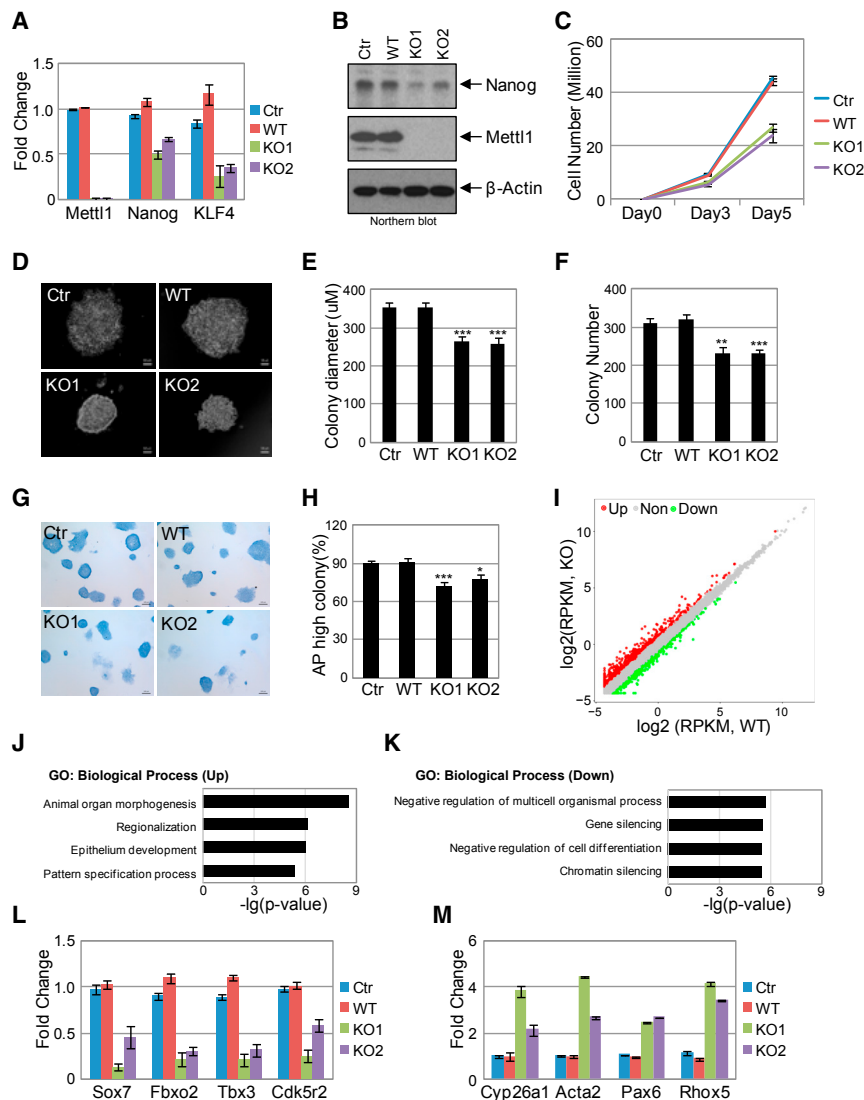


Figure 5. Mettl1 Is Required for mESCs Self-Renewal

(A) qRT-PCR analysis of mRNA expression in Mettl1 KO and control cells with indicated primers. The error bars represent mean $\pm$ SEM; $n = 3$. (B) Western blot with indicated antibodies. (C) Cellular proliferation in the Mettl1 KO and control cells. 10^5 cells were seeded in 100-mm tissue culture dishes and then the cell numbers were counted at day 3 and day 5. $n = 3$. (D–F) Colony-formation assay of Mettl1 KO and control cells. 500 cells were seeded in 6-well plate; one week later, the colony size and number were measured. (D) The representative images of the colonies. The scale bar represents 50 μ M. (E) Colony diameters. $n = 8$; *** $p < 0.001$. (F) Colony numbers. $n = 6$; ** $p < 0.01$; *** $p < 0.001$. (G and H) AP staining of the Mettl1 KO and control cells. The scale bar represents 100 μ M. (G) Representative images; (H) percentage of AP-positive colonies. $n = 10$; * $p < 0.05$; *** $p < 0.001$. (I) Scatterplot of mRNA expression using the RNA-seq data from the Mettl1 KO and control cells. (J and K) Gene ontology analysis of the upregulated (J) and downregulated (K) genes upon Mettl1 KO in mESCs. (L and M) qRT-PCR analysis of mRNAs with decreased (L) or increased (M) expression in Mettl1 KO and control cells with indicated primers. The error bars represent mean $\pm$ SEM; $n = 3$.

Mettl1 Deficiency Disrupts ESC Differentiation to Neural Lineages

We next explored the requirement of Mettl1-mediated m⁷G tRNA methylome in mESC differentiation. First, we induced spontaneous differentiation of mESCs by embryoid body (EB) formation assay (Figure 6A), in which the mESCs differentiate into derivatives of all three embryonic germ layers. The Mettl1 KO cells and the control cells were induced to differentiate for 6 days, and the relative expression of a panel of different lineage markers was measured by qRT-PCR. Interestingly, we found that depletion of Mettl1 selectively increased the expression of mesoderm and endoderm lineage markers, and the expression of ectoderm markers was decreased or remained unchanged in the Mettl1 KO cells (Figure 6A); this suggested that Mettl1 depletion might disrupt the balance of ESCs differentiation. Transcriptome profiling of differentiated Mettl1 KO and control cells revealed that depletion of Mettl1 resulted in the increased expression of 466 genes, most of which are involved in embryonic

development, such as cardiovascular system development (Figures 6B and 6C; Table S4). On the other hand, Mettl1 KO resulted in the downregulation of 218 genes that are associated with the reproductive process and fertilization (Figure 6C; Table S4). Overall, our data revealed that Mettl1 deficiency leads to skewed differentiation toward the mesoderm and endoderm lineages and uncovers an essential function of Mettl1-mediated m⁷G tRNA methylome in mESC fate determination.

Mutations in the Mettl1-interacting partner WDR4 result in defective tRNA m⁷G modification and severe brain malformations in human patients. Considering that Mettl1 depletion favored the differentiation of mESCs to mesoderm and endoderm at the expense of ectoderm that gives rise to the entire nervous system, we specifically induced differentiation to the neural lineage and analyzed the effect of Mettl1 deficiency to determine whether Mettl1 KO impaired the neural differentiation potential of mESCs. For this purpose, the Mettl1 KO and control cells were cultured in the N2B27 medium to induce differentiation toward neural lineages for 6 days (Ying et al., 2003; Figure 6D). We measured relative expression of a selection of neuronal marker genes by qRT-PCR. Our data revealed the Mettl1 depletion resulted in a striking decrease in the expression of neural lineage genes (Figure 6D), supporting that Mettl1 KO cells have severely compromised neural differentiation. In addition, overexpression

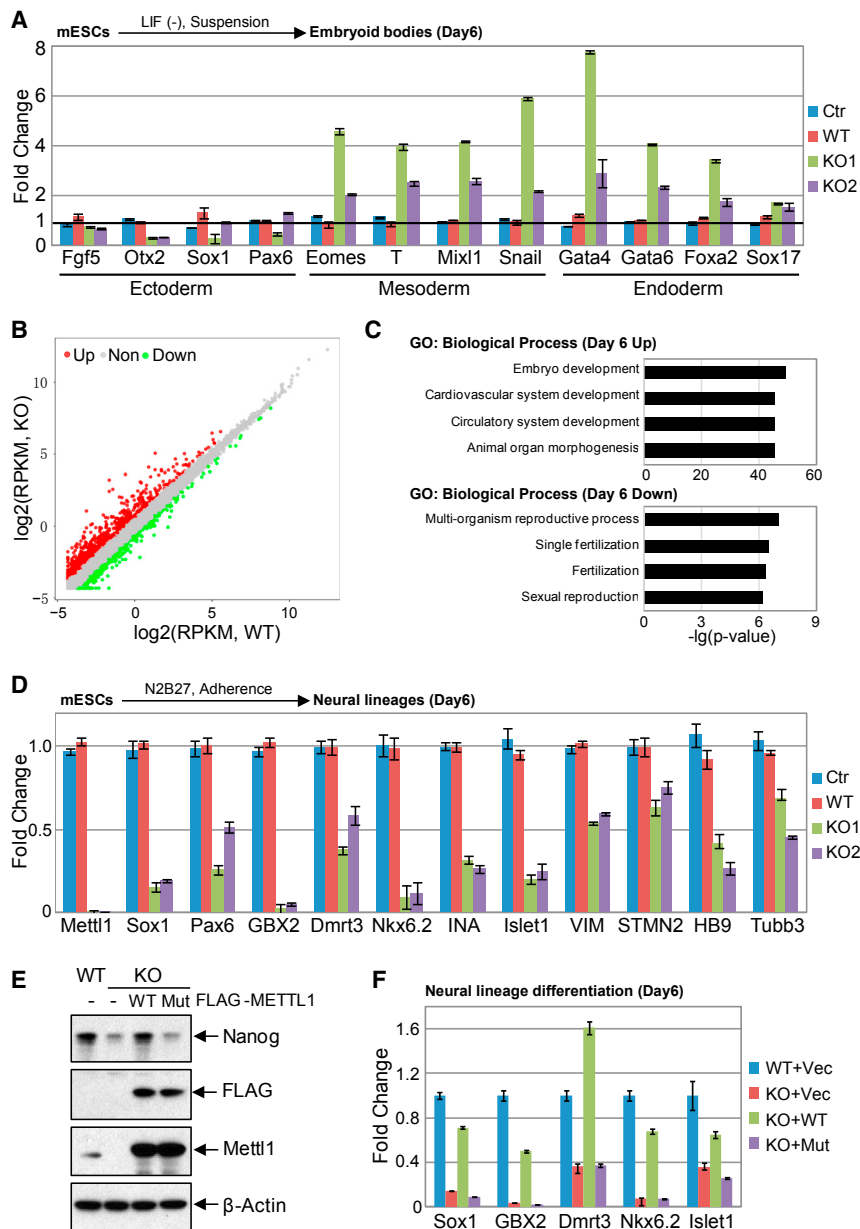


Figure 6. Mettl1 KO Disrupts Balance of ESC Differentiation and Impairs Neural Lineage Differentiation

(A) Upper panel: schematic diagram of ESC embryoid body differentiation procedure. Lower panel: qRT-PCR analysis of mRNA expression in Mettl1 KO and control cells (after differentiation for 6 days) with indicated primers is shown. The error bars represent mean $\pm$ SEM; $n = 3$.

(B) Scatterplot of mRNA expression using the RNA-seq data from the Mettl1 KO and control embryonic bodies (differentiated for 6 days).

(C) Gene ontology analysis of functional enrichment in biological process using the differentially expression genes between the day 6 Mettl1 KO and control embryoid bodies.

(D) Upper panel: schematic diagram of ESC neural lineage differentiation procedure. Lower panel: qRT-PCR analysis of mRNA expression in Mettl1 KO and control cells (after differentiation for 6 days) with indicated primers is shown. The error bars represent mean $\pm$ SEM; $n = 3$.

(E) Western blot of the METTL1 rescue samples with indicated antibodies.

(F) qRT-PCR analysis of mRNA expression in METTL1 rescue samples (after differentiation for 6 days) with indicated primers. The error bars represent mean $\pm$ SEM; $n = 3$.

ylated tRNAs and the longer CDS and UTR in the neural lineage genes might explain why neural lineage genes are more affected by the loss of m⁷G tRNA modification. Overall, our data uncover that Mettl1 deficiency disrupted the balance of ESCs differentiation and impaired differentiation to the neural lineage.

Wdr4 Deficiency Abolishes m⁷G tRNA Modification and Impairs ESC Proliferation and Neural Differentiation

To further study the role of m⁷G tRNA methylation in regulation of stem cell pluripotency and differentiation, we generated the Wdr4 KO mESCs (Figures 7A

and 7B). Removal of the exons 4–6 of Wdr4 gene by CRISPR/Cas9 resulted in strong decrease of Wdr4 mRNA expression (Figure 7C). Our Northwestern blot showed that KO of Wdr4 resulted in complete loss of m⁷G modification in tRNAs (Figure 7D). Anti-m⁷G MeRIP and chemical-mediated cleavage assay further confirmed the critical function of Wdr4 in mediating the m⁷G tRNA modification in mESCs (Figures 7E and 7F). We next studied the role of Wdr4 in the regulation of mESC proliferation and differentiation. Similar to Mettl1 KO, depletion of Wdr4 resulted in the slower proliferation of mESCs (Figure 7G). Most importantly, when induced for neural lineage differentiation, the WDR4 KO cells showed decreased expression of neural differentiation markers (Figure 7H), suggesting that depletion of WDR4 impairs the neural lineage differentiation capacity of

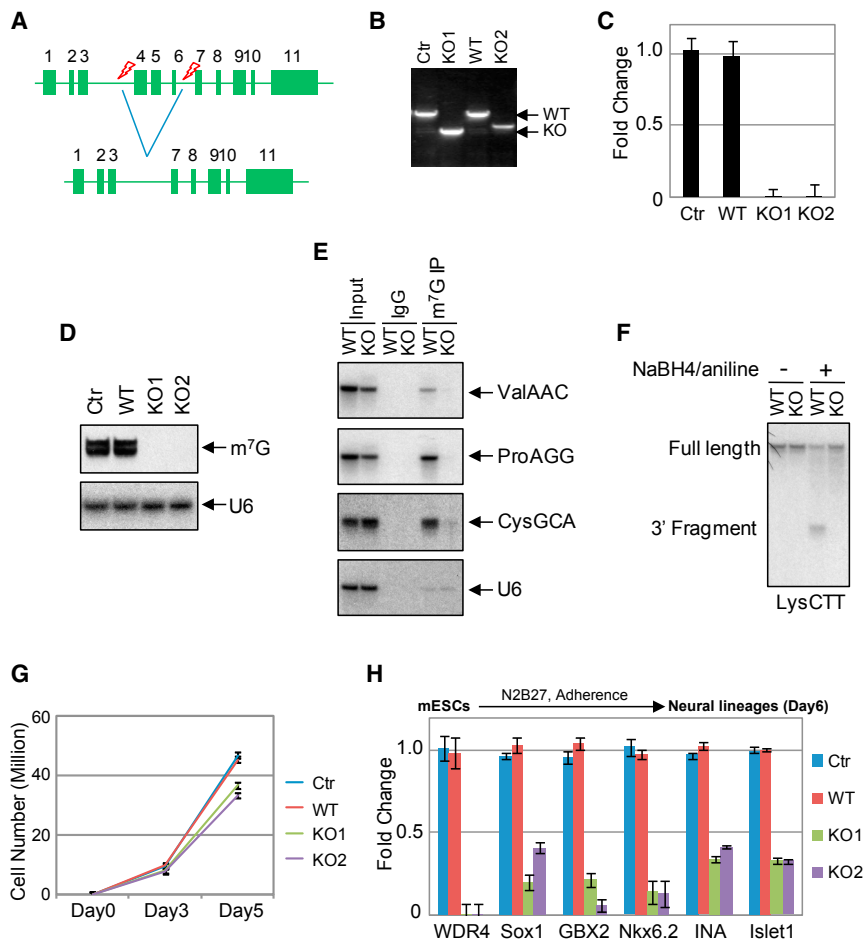


Figure 7. *Wdr4* Knockdown Abolishes m⁷G tRNA Modification and Impairs ESC Proliferation and Neural Differentiation

(A) Schematic diagram of CRISPR-mediated KO of *Wdr4* expression in mESCs.

(B) PCR to confirm the KO of *Wdr4* in mESCs using genome DNA samples from the KO and control cells.

(C) qRT-PCR analysis of *Wdr4* mRNA expression in KO and control cells. The error bars represent mean $\pm$ SEM; n = 3.

(D) Anti-m⁷G Northwestern blot of m⁷G modifications. RNA samples were separated with TBE-UREA gel, when transferred to nylon membrane for western blotting using anti-m⁷G antibody. Northern blot of U6 snRNA was used as a loading control.

(E) Small RNA northern blot of the anti-m⁷G IP products with indicated probes.

(F) Northern blot of the chemical-treated RNA samples with indicated probes.

(G) Cellular proliferation in the *Wdr4* and control cells. 10⁵ cells were seeded in 100-mm tissue culture dishes and then the cell numbers were counted at days 3 and 5.

(H) Upper panel: schematic diagram of ESC neural lineage differentiation procedure. Lower panel: qRT-PCR analysis of mRNA expression in *Mettl1* KO and control cells (after differentiation for 6 days) with indicated primers is shown. The error bars represent mean $\pm$ SEM; n = 3.

mESCs. Taken together, our data support the essential roles of the *Mettl1*- and *Wdr4*-mediated m⁷G tRNA modification in regulation of mESCs self-renewal and differentiation.

DISCUSSION

This study developed two methods, including m⁷G MeRIP-seq and TRAC-seq, to reveal the mammalian m⁷G tRNA methylome at single-nucleotide resolution. We find a specific subset of 22 tRNAs that are m⁷G modified and contain a RAGGU sequence motif in the variable loop. Unlike in yeast, where m⁷G is dispensable for growth under normal conditions, our work reveals an essential role for m⁷G tRNA modification for ESC self-renewal and neural differentiation, and this is most likely explained by the more complex m⁷G tRNA modification landscape in mammals and a particular requirement of the m⁷G-modified tRNAs for the efficient translation of mRNAs involved in mitosis and brain development. This conclusion is based on loss-of-function studies of the components of the m⁷G methyltransferase complex *Mettl1* and *Wdr4*, where we observe a complete loss of m⁷G tRNA modification. Ribosome profiling identifies widespread effects on mRNA translation, especially for mRNAs related to the cell cycle and those controlling forebrain morphology that are less efficiently translated in *Mettl1* KO

conditions. Moreover, we found *Mettl1* or *Wdr4* KO ESCs have defects in cell differentiation, in particular showing compromised ability to give rise to neural lineages.

The antibody-based methylated RNA immunoprecipitation coupled with next-generation sequencing (MeRIP-seq) technique is a powerful strategy for transcriptome-wide mapping of RNA modifications in mRNAs (Dominissini et al., 2013; Zhao et al., 2017). However, there was no reported global MeRIP-seq-mediated profiling of tRNA modifications because the extensive tRNA modifications interfere with reverse transcription during cDNA library construction. Recently, two independent groups developed efficient tRNA sequencing methods (DM-tRNA-seq and ARM-seq). Both methods use the bacterial demethylase AlkB to remove tRNA methylation to facilitate efficient reverse transcription (Cozen et al., 2015; Zheng et al., 2015). These tRNA sequencing methods served as a basis for our development of the tRNA MeRIP-seq technique for the global profiling of m⁷G tRNA modification. We combined the m⁷G MeRIP and the AlkB-mediated tRNA sequencing approach to develop the m⁷G tRNA MeRIP-seq method (Figure 1), which allowed us to profile the global m⁷G tRNA methylome to study the molecular mechanisms underlying the dysregulated m⁷G tRNA modifications in diseases. Whereas we focused here on m⁷G because of the severe phenotypes associated with altered

m⁷G modifications in human patients, we anticipate that adaptation of our method using antibodies that recognize other RNA modifications will be useful to interrogate and map the global landscape of other tRNA modifications. To confirm and extend our findings from the m⁷G tRNA MeRIP-seq method, we further developed an antibody-independent, tRNA reduction and cleavage-sequencing (TRAC-seq) method. In TRAC-seq, the small RNA-enriched samples are first treated with AlkB, which can remove other dominant methylation marks, but not m⁷G modification; the RNAs are then subsequently treated with sodium borohydride followed by aniline to induce the site-specific cleavage of tRNAs at the m⁷G residue (Wintermeyer and Zachau, 1975; Zueva et al., 1985). The 3' fragment released by this chemical cleavage carries a 5' phosphate that is compatible with linker ligation, thereby enabling cDNA library preparation for next-generation sequencing. m⁷G TRAC-seq specifically identified the chemical-induced cleavage sites, facilitating the global mapping of m⁷G modification at single-nucleotide resolution (Figure 2).

Our m⁷G tRNA MeRIP-seq and m⁷G TRAC-seq methods identified 22 m⁷G-modified tRNAs in mESCs that possess a RAGGU motif, revealing much more widespread m⁷G tRNA modifications in mammals than are known for yeast. The requirement of the RAGGU sequence motif could explain the larger number of m⁷G-modified tRNAs in mammals than in yeast: (1) the mouse genome contains a larger repertoire of tRNAs, and some of the tRNAs present in mouse, such as GlyAAC, contain a RAGGU motif and therefore are modified (Figure 3B). (2) The sequences of certain tRNAs are slightly different between different species during evolution; for example, the mouse ArgTCT tRNA has AAGGU motif in the V-loop, and the corresponding sequence in yeast ArgTCT tRNA is AAGAU, such that it lacks the "G" that is modified in mouse tRNA. Interestingly, most of the mouse mitochondrial tRNAs do not have the RAGGU motif, and concomitantly, we did not detect m⁷G modification in mitochondrial tRNAs (data not shown). Analysis of all mouse tRNAs in the database revealed that 2 additional tRNAs contain the RAGGU, SerGGA contains AAGGU and HisATG contains "GAGGU," but are not included in our list of 22 tRNA because they were not detected in mESCs yet based on our findings might be expected to be m⁷G modified in other cell types where they are expressed.

Polysome profiling and ribosome footprinting revealed that depletion of Mettl1-mediated m⁷G tRNA modifications resulted in ribosome pausing at m⁷G tRNA-dependent codons. Interestingly, the genes that are translationally regulated by the Mettl1 are enriched in cell cycle genes and the genes associated with brain abnormality. Consistent with the gene ontology analysis, Mettl1 KO in mESCs increased G2 phase percentage and resulted in slower proliferation and impaired colony-formation ability. AP staining and gene expression profiling revealed that the Mettl1-depleted mESCs undergo partial spontaneous differentiation. Overall, the above data revealed that Mettl1-mediated m⁷G tRNA modification is essential to maintain ESC self-renewal. Most importantly, when induced for spontaneous differentiation, the Mettl1 KO cells tend to differentiate into mesoderm and endoderm lineages, but not ectoderm. Neural

specific differentiation of mESCs further confirmed that Mettl1-mediated tRNA m⁷G modification is essential for the proper expression of neural lineage genes. Considering that (1) Mettl1 depletion decreases the translation of genes associated with abnormality of forebrain morphology, cerebrum, and skull size; (2) Mettl1 KO in mESCs results in impaired neural differentiation; and (3) mutation of WDR4 causes impaired tRNA m⁷G modification and microcephalic primordial dwarfism with neural defects (Shaheen et al., 2015; Trimouille et al., 2018), our data provide strong evidence for the molecular and cellular requirement for m⁷G tRNA modifications that is likely responsible for the pathogenesis of this human disease. It will be of substantial interest to apply m⁷G MeRIP-seq and/or TRAC-seq to explore changes in the tRNA methylome and links with human developmental disorders, including microcephalic primordial dwarfism with neural defects as well as Down syndrome, to better understand these tantalizing links between the tRNA methylome and disease.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

Supplemental Information includes six figures and five tables and can be found with this article online at <https://doi.org/10.1016/j.molcel.2018.06.001>.

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AUTHOR CONTRIBUTIONS

S.L. designed and performed most of the experiments. Q.L. performed most of the bioinformatics analysis with input from S.L. V.S.L. optimized the RNA chemical reduction and cleavage conditions. V.S.L. and J.W.S. helped conceive of the RNA cleavage and sequencing method. J.C. performed the polysome profiling assay. J.W.S. and R.I.G. designed and supervised the project. S.L., Q.L., and R.I.G. wrote the paper with input from other authors.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Rabbit polyclonal anti-Mettl1	Proteintech	Cat# 14994-1-AP
Rabbit polyclonal anti-beta-Acin	Abcam	Cat# ab8227
Rabbit polyclonal anti-Nanog	Abcam	Cat# ab70482
Mouse monoclonal 7-methylguanosine (m ⁷ G)	MBL International	Cat# RN017M
Mouse monoclonal anti-FLAG	Sigma	Cat# A8592
Bacterial and Virus Strains		
GC10 chemically competent cells	Genesee Scientific	Cat# 42-657
BL21(DE3) Competent <i>E. coli</i>	New England Biolabs	Cat# C25271
Chemicals, Peptides, and Recombinant Proteins		
DMEM	GIBCO	Cat#11965092
DMEM/F12	GIBCO	Cat#11330032
Neurobasal medium	GIBCO	Cat#21103049
Opti-MEM I Reduced Serum Medium	GIBCO	Cat# 31985070
MEM Non-Essential Amino Acids Solution (NEAA)	GIBCO	Cat#11140050
L-Glutamine	GIBCO	Cat# 25030081
Sodium Pyruvate	GIBCO	Cat# 11360070
Penicillin-Streptomycin	GIBCO	Cat# 15140163
B-27 Supplement	GIBCO	Cat# 12587010
N-2 Supplement	GIBCO	Cat# 17502048
Trypsin-EDTA	GIBCO	Cat# 15400054
DPBS	GIBCO	Cat# 14190250
Stem Cell Qualified Fetal Bovine Serum	Gemini	Cat# 100-125
Mouse Leukemia Inhibitory Factor (LIF)	Gemini	Cat# 400-495
Gelatin from porcine skin	Sigma	Cat# G1890
Formaldehyde solution	Sigma	Cat# 252549
Acetic acid glacial, ≥99.85%	Thermo Fisher Scientific	Cat# A38-212
2-Mercaptoethanol	Thermo Fisher Scientific	Cat# 60-24-2
Poly(2-hydroxyethyl methacrylate) (polyHEMA)	Sigma	Cat# 192066
ZnCl ₂	Sigma	Cat# Z0152
Trizma base	Sigma	Cat# T4661
EDTA	Sigma	Cat# 431788
NaCl	Sigma	Cat# S7653
KCl	Sigma	Cat# P9541
MgCl ₂	Sigma	Cat# M8266
Igepal CA-630	Sigma	Cat# I8896
Bovine Serum Albumin	Sigma	Cat# 05470
Isopropyl β-D-1-thiogalactopyranoside (IPTG)	Sigma	Cat# I5502
MES	Sigma	Cat# M3671
α-Ketoglutaric acid	Sigma	Cat# 75890
Ammonium iron(II) sulfate hexahydrate	Sigma	Cat# 215406
L-Ascorbic acid	Sigma	Cat# A92902
Sodium borohydride	Sigma	Cat# 452882
Aniline	Sigma	Cat# 242284

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Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
RNasin Ribonuclease Inhibitors	Promega	Cat# N2518
Lipofectamine 2000 Transfection Reagent	Invitrogen	Cat# 11668019
SuperScript II Reverse Transcriptase	Invitrogen	Cat# 18064014
SuperScript III Reverse Transcriptase	Invitrogen	Cat# 18080093
Platinum High Fidelity Taq DNA Polymerase	Invitrogen	Cat# 11304029
TRIzol Reagent	Invitrogen	Cat# 15596026
Critical Commercial Assays		
TruSeq Stranded mRNA Sample Prep Kits	Illumina	Cat# RS-122-2101
TruSeq Ribo Profile for Mammalian	Illumina	Cat# PHMR12126
Ribo-Zero Gold rRNA Removal Kit	Illumina	Cat# MRZG12324
NEBNext Small RNA Library Prep Kit	New England Biolabs	Cat# E7330S
Q5 Site-Directed Mutagenesis Kit	New England Biolabs	Cat# E0554
Fast SYBR Green Master Mix	Applied Biosystems	Cat# 4385612
PolyATtract mRNA Isolation System	Promega	Cat# Z5310
MinElute PCR Purification Kit (50)	QIAGEN	Cat# 28004
Oligo Clean & Concentrator	Zymo Research	Cat# D4061
Blue-color AP Staining Kit	SBI	Cat# AP100 B-1
Deposited Data		
RNA-seq, m ⁷ G-MeRIP-Seq, m ⁷ G-TRAC-Seq	This paper	GEO: GSE112670
Experimental Models: Cell Lines		
Mouse cell line: R1/E mESCs	ATCC	SCRC-1036
Oligonucleotides		
Oligonucleotides were listed in Table S5	This paper	N/A
Recombinant DNA		
pFLAG-CMV2 vector	Sigma	Cat# E7033
pET30a-AlkB	Addgene	Cat# 79050
pET30a-AlkB-D135S	Addgene	Cat# 79051
pX330	Addgene	Cat# 42230
Software and Algorithms		
trim_galore	https://www.bioinformatics.babraham.ac.uk/projects/trim_galore	N/A
Bowtie2	http://bowtie-bio.sourceforge.net	N/A
TopHat	https://ccb.jhu.edu/software/tophat/manual.shtml	N/A
Toppgene	https://toppgene.cchmc.org/	N/A
HTSeq	https://htseq.readthedocs.io/en/release_0.10.0/	N/A
edgeR	http://bioconductor.org/packages/release/bioc/html/edgeR.html	N/A
Bedtools	http://bedtools.readthedocs.io/en/latest/	N/A

CONTACT FOR REAGENT AND RESOURCE SHARING

Further queries and reagent requests may be directed and will be fulfilled by the lead contact, Richard I. Gregory (rgregory@enders.tch.harvard.edu).

EXPERIMENTAL MODEL AND SUBJECT DETAILS**Cell Culture**

Mouse R1/E ESCs (gift from Dr. Lizi Wu) were cultured in either Serum/LIF medium [DMEM (GIBCO) with 1000u/ml mLIF (Gemini), 15% stem cell FBS (Gemini), 1X Sodium Pyruvate (GIBCO), 1X NEAA (GIBCO), 1X L-glutamine (GIBCO), 50μM 2-mercaptoethanol

(ThermoFisher) and 1% Penicillin-Streptomycin (GIBCO)] in a 5% CO₂ cell culture incubator at 37°C. Cells were cultured in feeder free condition, and the cultured dishes were coated with 0.1% Gelatin for 30 min at 37°C to enhance cell attachment. For cell passage, cells were first washed with PBS for twice. Trypsin (0.1% in PBS) was added to the dishes, which were then incubate at 37°C for 3 min, and Serum/LIF medium above was used to inactive trypsin. After collected by centrifuge at 1000 g for 3 min, cells were counted and split for different purposes. To knockout Mettl1 or WDR4 with CRISPR technique, sgRNAs targeting Mettl1 or WDR4 were cloned into pX330 vectors (Cong et al., 2013). Then the vectors were transfected into mESCs with Lipofectamine 2000 (Invitrogen) together with a puromycin resistant plasmid. The second day after transfection, cells were seeded in low density and selected with 2.5ug/ml puromycin for 24 hr. One week later, single colonies were screened with PCR to identify the knockout clones. The sgRNAs sequences and primer sequences were listed in Table S5.

METHOD DETAILS

Plasmid Construction

For preparation of Mettl1 or WDR4 sgRNA vector for CRISPR knockout, sgRNAs were designed using CRISPR design tool (<http://crispr.mit.edu>). Sense and antisense sgRNA oligos were annealed and then ligated into BbsI site of pX330 vector following the instructions from Addgene (<http://www.addgene.org/crispr/zhang>). For cloning of METTL1 expression, the full-length METTL1 cDNA was first PCR amplified from the 293T cDNA with primers that contain the NotI (forward primer) and BglII (reverse primer) sites. Then the PCR product was gel purified and digested with NotI and BglII at 37°C overnight. After that the digested PCR product was purified again and ligated into the NotI/BglII cut pFLAG-CMV2 vector. The METTL1 catalytic dead mutant (aa160-163, LFPD to AFPA) was generated using the Q5 Site-Directed Mutagenesis Kit (New England Biolabs).

mESCs Self-Renewal and Differentiation

To determine self-renewal capacities of the Mettl1 or Wdr4 knockout and control mESCs, 10⁵ cells were seeded in 100 mm tissue culture dishes and then the cell numbers were counted at Day3 and Day5. For colony formation assay, 500 cells were seeded in 6 well plate, the colony size and number were measured one week later. Alkaline phosphatase staining was performed with Blue-Color AP Staining Kit (System Biosciences) following the manufacturer's instructions. For spontaneous differentiation of mESCs, the cell culture dishes were first coated with Poly(2-hydroxyethyl methacrylate) (polyHEMA) to prevent cell attachment. Briefly, 2ml polyHEMA (Sigma) (20mg/ml in 75% ethanol) was added into 100 mm dishes, which were dried in cell culture hood for more than 4 hr. Then 5 X 10⁶ mESCs were cultured with LIF (-) medium in the Poly(2-hydroxyethyl methacrylate) coated dishes to induce embryoid body formation for 6 days. To induce the neural lineage differentiation of mESCs, 5 X 10⁵ mESCs were cultured in N2B27 medium for 6 days using a previously established protocol (Ying et al., 2003). Briefly, 60mm dishes were first coated with 0.1% Gelatin, then 5 X 10⁵ mESCs were cultured in 60 mm dishes using the N2B27 medium [1:1 mixture of DMEM/F12 (GIBCO) supplemented with N2 and Neurobasal medium supplemented with B27 (GIBCO)].

RNA Isolation, Reverse Transcription, and qRT-PCR

Total RNA samples were isolated with Trizol following the manufacturer's instructions. Briefly, cells were washed with PBS twice, then 1ml Trizol were added to dishes for RNA isolation. The concentrations of isolated RNA samples were measured by Nanodrop. Reverse transcription was performed with SuperScript III Reverse Transcriptase (Invitrogen) using 2μg total RNA in a 20ul reaction system. Random hexamers were used for the reverse transcription. After reverse transcription, the cDNA samples were diluted 1:10 for the analysis of gene expression using qRT-PCR. qRT-PCR was performed using SYBR Green PCR Master Mix with the Step One Real-Time PCR System (Applied Biosystems). The primers used for qRT-PCR assays were listed in Table S5.

Northern Blot, Northwestern Blot, and Western Blot

Northern blot and western blot were performed as previously described (Lin and Gregory, 2015). Briefly, for Northern blotting of tRNAs or U6 snoRNA, 2ug total RNA samples were mixed with 2X TBE loading buffer (Bio-Rad) and incubated at 95°C for 5 min, then the samples were loaded into 10% TBE-UREA (Bio-Rad) gel to separate the RNAs by molecular weight. 5% TBE-UREA gel was used for the Northern blot of 18S rRNA. After electrophoresis, the RNAs were transferred onto a positive charged nylon membrane. The RNAs on the membrane were crosslinked with UV and then the membrane was blotted with radioactive labeled probes against tRNAs or 18S rRNA, or U6 snRNA. The probe sequences were listed in Table S5. For Northwestern blotting, the RNA containing membranes were blotted with anti-m⁷G antibody (MBL International, #RN017M), and then the m⁷G signals were detected following the western blot protocol described previously (Lin and Gregory, 2015).

tRNA Aminoacylation Analysis

tRNA aminoacylation was determined by acid urea polyacrylamide gel electrophoresis (acid urea PAGE) followed by Northern blot method as previously described (Janssen et al., 2012; Köhrer and Rajbhandary, 2008). Briefly, RNA samples were isolated with Trizol and treated with 0.1 M Tris PH9.5 at 37°C for 1 hr, then the treated and control RNA samples were precipitated with 2.5 volume of ethanol and 1/10 volume of 3M sodium acetate (pH5.2), and resuspended in 10 mM sodium acetate pH 5.0. 2ug RNA samples were loaded into acid (pH5.0) urea polyacrylamide gel, then the RNAs were separated by electrophoresis using 0.1 M sodium acetate

(pH 5.0) as electrophoresis buffer. After electrophoresis, the RNAs in the gel were transferred onto a positive charged nylon membrane. The RNAs on the membrane were crosslinked with UV and then the membrane was blotted with radioactive labeled probes against m⁷G modified tRNAs. The probe sequences were listed in [Table S5](#).

Recombinant Protein Purification

Recombinant wild-type and D135S AlkB proteins were purified as previous described ([Lin and Gregory, 2015](#)). pET30a-AlkB and pET30a-AlkB-D135S were gifts from Tao Pan (Addgene plasmid # 79050 and #79051) ([Zheng et al., 2015](#)). pET30a-AlkB and pET30a-AlkB-D135S were transformed into BL21 bacteria for induced expression of recombinant proteins. Bacteria were inoculated and cultured in LB medium at 37°C. Recombinant wild-type and D135S AlkB protein expressions were induced in BL21 bacteria (OD 0.4-0.6) using 0.5mM IPTG at 20°C overnight. Then the bacteria were collected and lyzed by sonication, centrifuged at 15000rpm at 4°C for 60 min. The supernatant were collected for the purification of recombinant proteins using Ni-NTA Agarose (QIAGEN) following the manufacturer's instructions.

m⁷G tRNA meRIP-Seq

For m⁷G tRNA meRIP-Seq, small RNAs (< 200nt) were first purified using the mirVana miRNA Isolation Kit (Thermo Fisher Scientific). Then anti-m⁷G meRIP was performed on the small RNA by modifying the previously established mRNA meRIP protocol ([Dominissini et al., 2013](#); [Lin et al., 2016](#)). Briefly, 75 µg small RNAs were incubated with 7.5 µg anti-m⁷G antibody (MBL International, #RN017M) for 2 hr, and then 100 µl pre-washed Protein A/G Magnetic Beads (Thermo Fisher Scientific, #88802) were added to purify the m⁷G modified small RNAs. After extensive wash, the beads bound RNAs were extracted with Trizol reagent following the manufacturer's instructions. The precipitated small RNAs and input small RNAs were treated with purified recombinant wild-type and D135S AlkB proteins to remove dominant methylations as previously described ([Zheng et al., 2015](#)). Briefly, the precipitated small RNAs and 1 µg input small RNAs were treated with 80 pmol wtAlkB and 160 pmol D135S AlkB mutant in a 100 µl demethylation reaction [300 mM KCl, 2 mM MgCl₂, 50 µM of (NH₄)₂Fe(SO₄)₂·6H₂O, 300 µM 2-ketoglutarate (2-KG), 2 mM L-ascorbic acid, 50 µg/mL BSA, 50 mM MES buffer (pH 5.0)]. After incubation at room temperature for 2 hr, the reaction was quenched with at final concentration of 5 mM EDTA. Then the treated RNAs were purified by phenol-chloroform extraction and ethanol precipitation. After that, small RNA libraries were constructed with NEBNext Small RNA Library Prep Set (New England Biolabs) ([Cozen et al., 2015](#)). The cDNA libraries were sequenced with an Illumina Nextseq 500.

m⁷G TRAC-Seq

For m⁷G TRAC-Seq, isolated small RNAs were first treated with recombinant wild-type and D135S AlkB proteins to remove the dominant methylations on RNAs. 2.5 µg AlkB-treated RNAs were treated with 0.1M NaBH₄ for 30 min on ice at dark in the presence of 1 mM free m⁷GTP as methylation carrier. Then the RNAs were precipitated with 0.1 volume 3M sodium acetate (pH5.2) and 2.5 volume cold ethanol at -20°C for at least 2 hr. After precipitation, the NaBH₄-treated RNAs were subsequently treated with aniline-acetate solution (H₂O: glacial acetate acid:aniline, 7:3:1) at room temperature at dark for 2 hr to induce the site specific cleavage ([Wintermeyer and Zachau, 1975](#); [Zueva et al., 1985](#)). The RNA samples were purified by ethanol precipitation and used for cDNA library construction, followed by sequencing with Illumina Nextseq 500.

Ribosome Footprinting

Ribosome footprinting was performed using the TruSeq Ribo Profile kit from Illumina. Briefly, cells were treated with cycloheximide, then the ribosome protected fragments were generated by Nuclease digestion. Ribo-Zero Gold kit (Illumina) was used to remove rRNA. The purified RPF and fragmented input RNA samples were subjected to end repair, 3' adaptor ligation, reverse transcription, cDNA circularization and library amplification following the manufacturer's instructions.

Ribo-Seq Data Analysis

Adaptor sequences were trimmed and low quality sequences (Q30) were discarded using trim_galore (https://www.bioinformatics.babraham.ac.uk/projects/trim_galore). The sequences with length at less 25 nt were further aligned to mouse rRNA and tRNA sequence index constructed from Ensembl (Release 91) (https://useast.ensembl.org/Mus_musculus/Info/Index) and GtRNAdb databases ([Chan and Lowe, 2016](#)), respectively. The remaining reads were aligned to the UCSC canonical known gene transcripts of the mouse reference genome (mm10) using Bowtie ([Langmead et al., 2009](#)) with a maximum one mismatch allowed. The uniquely mapped reads with mapping quality ≥ 20 were used for further analysis.

For codon occupancy analysis, 28-31 nt reads translated in the 0-frame of CDS were used to infer the A site codon with an offset of 15 nt from 5' end of the read ([Nedialkova and Leidel, 2015](#)). The footprinting reads mapped in the first 15 codon of CDS were not taken into account to exclude the influence of sample preparation. The codon occupancy of an A-site was normalized by its basal occupancy which is computed as the average codon occupancy of these sites in +1, +2 and +3 positions downstream of the A site ([Nedialkova and Leidel, 2015](#)).

For ribosome occupancy analysis, the translation efficiency (TE) was calculated by dividing the CDS footprinting abundance (excluding the starting 15 codons) by its mRNA abundance. The genes with 1.5 fold change of TE were considered as the differential translation genes ([Zinshteyn and Gilbert, 2013](#)).

tRNA, tRNA-IP and Alkb Treated tRNA Sequencing Data Analysis

The mouse tRNA sequences (mm10) were downloaded from GtRNAdb database (Chan and Lowe, 2016). The microRNA sequences and other small non-coding RNA sequences (such as snRNA, snoRNA and rRNA) were retrieved from miRBase database (Release 21) (<http://www.mirbase.org/>) and Ensembl noncoding database (Release 91) (https://useast.ensembl.org/Mus_musculus/Info/Index), respectively. The mRNA and long noncoding RNA sequences were downloaded from RefSeq database and NONCODE database (Version 5, <http://www.noncode.org>), respectively. To further process the tRNA sequence, the sequences containing the tRNA genes and 100 bp at both upstream and downstream of tRNA genes were extracted for precursor tRNA genes. For mature tRNA sequences, the introns were removed and “CCA” was added to the 3' end of the tRNA gene sequences.

After adaptor trimming and low quality filtering (Q30), the clean sequencing data were first mapped to the tRNA sequences using Bowtie2 (<http://bowtie-bio.sourceforge.net>). Then read count for each tRNA was calculated using the method in Cozen et al. (2015). The read counts of tRNA are then normalized as reads per million (RPM) by the total number of small RNA reads that are matched to tRNAs in each sample. For sequence mapping statistic plots, the clean reads were mapped to different types of RNA libraries with the orders: tRNA, miRNA, snRNA, snoRNA, rRNA, other sncRNA, mRNA and long non-coding RNAs. The non-mapped reads were lastly mapped to mouse reference genome (mm10).

RNA-Seq Expression Analysis of ESC Differentiation

For RNA-seq data of ESC differentiation samples (D0, D6) with METTL1 Knockdown, each time point has two replicates. After adaptor trimming and low quality sequence filtering (Q30), the clean reads were mapped to mouse reference genome (mm10) using Tophat2 (Kim et al., 2013). The raw counts of reads mapping to the GENCODE genes (v19) (Harrow et al., 2012) were computed by HTSeq with ‘union’ overlap mode (Anders et al., 2015). The raw counts were then normalized as Reads per kilo base per million mapped reads (RPKM) using edgeR (Robinson et al., 2010). The genes with 1.5 fold change and FDR < 0.05 were considered as the differential expressed genes.

tRNA Chemical Sequencing Data Analysis

For tRNA chemical sequencing data, clean reads were mapped to the mouse mature tRNA sequences generated previously using Bowtie with a maximum of two mismatches allowed (Langmead et al., 2009). The alignments were then processed to record the read depth of each site and the number of reads starting at that position using Bedtools (<http://bedtools.readthedocs.io/en/latest/>). We defined the cleavage ratio of site i as the ratio between the number of reads starting at site i and the read depth of site i (Schwartz et al., 2014). Then, the cleavage score of site was calculated as:

$$Cleavage\ score_i = \frac{\log 2(Cleavage\ ratio_{treat})}{\log 2(Cleavage\ ratio_{non-treat})}$$

The positions with a cleavage score > 6 and the cleavage ratio > 0.1 in both samples were considered as the candidate m⁷G sites.

Gene Ontology and Disease Ontology Analysis

Gene Ontology and Disease Ontology analysis of differential expressed genes and differential translation genes were conducted using the ToppFunn module of the ToppGene Suite (<https://toppgene.cchmc.org/enrichment.jsp>). The ontology terms with Benjamini Hochberg adjusted p value (FDR) < 0.05 were considered as significantly enriched.

QUANTIFICATION AND STATISTICAL ANALYSIS

Quantification and statistical analysis methods were described in individual method sections and Figure legends.

DATA AND SOFTWARE AVAILABILITY

Data Resource

The accession number for the RNA-seq, m⁷G MeRIP-Seq and TRAC-Seq data reported in this paper is GEO: GSE112670.